

Part II: Universal Measurable Estimators for ADAPT-VQE Gradient Selection

Steps 0–4 — methods, results and reproduction

Part II working note

September 2026

Abstract

We implemented Steps 0–4 of the Part II plan on the nine Part I benchmark cases.

- **Step 0.** Each measurement context in fact reads out a maximal group of 2^n commuting Pauli products. We built the tools to exploit this: measurement circuits, the moments of every group from one transform, and exact shot-level sampling. All are validated against dense linear algebra and against Part I.
- **Step 1.** The II-0 baseline runs Part I’s estimator as a shot-level experiment, with pairwise elimination, radius schedule $\gamma = 0.9$ and top-up allocation. Its median cost is 62–83% below Part I’s M3 as run. It made 5 wrong selections in 1,100 trials, all on the tied H₄ HF rows.
- **Step 2.** 58–75% of the Pauli products in $\mathcal{B}_0$ are measured by at least two contexts, so coefficient splitting is possible everywhere.
- **Step 3.** With exact covariances, splitting (II-A) lowers the M3 cost by 38–80% across all cases. Ghost Pauli products (II-B) add up to 10 points. Redesign at every elimination breakpoint (oracle II-D) adds up to 19 points, but only under top-up accounting.
- **Prior transfer.** A design fitted to Hartree–Fock covariances and applied to the correlated state is 66–69% *more* expensive than II-0.
- **Step 4.** The design is learned from the shots and the confidence radii are estimated too, so the run is fully non-oracle. It relies on cross-fitting, held-out radii and allocation, and a minimum number of shots per fold before a context is learned. On the two H₄ CISD rows and on LiH, II-A then costs 36–43% less than II-0 run the same way, with no wrong selections in 500 trials. That is 75–84% of the online oracle ceiling. The HF prior on its own is unreliable (85–99% correct, heavy tails), and no prior improved on the data alone.

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1 Problem and conventions

At one ADAPT-VQE step, with state $|\psi\rangle$ and pool $\{G_i\}_{i=1}^N$, the task is to identify $i^* = \arg \max_i |g_i|$, where

$$g_i = \langle C_i \rangle_\psi, \quad C_i = [H, G_i] = \sum_\ell A_{i\ell} P_\ell,$$

using as few *context-shots* as possible. One context-shot is one state preparation plus one measurement in one fully commuting (FC) context.

Cases. The nine Part I cases, rebuilt with the current Part I pipeline:

- systems: H₄ square with side 1.0 and 2.0 Å; LiH at $R = 3.0$ Å; H₂O at $R_{\text{OH}} = 0.9572$ and 1.75 Å;
- states: HF and CISD (LiH HF only);
- STO-3G, Jordan–Wigner, spin-orbital UCCSD pool ($N = 26, 92, 140$ on 8, 12, 14 qubits);
- Part I’s validation gate, and Part I’s deterministic first-fit FC partition of $\mathcal{B}_0 = \bigcup_i \text{supp}(C_i)$ into parent contexts $\mathcal{M}_\alpha$.

The rebuilt $|g_i|$ agree with Part I’s committed values to 3×10^{-10} . Three H₄ rows differ in the sign of some g_i . This is an orbital-phase convention, and every quantity reported here is invariant under it.

Confidence and radii. A confidence radius is $r = z_\delta \epsilon$, with ϵ the one-standard-error target and $z_\delta = \Phi^{-1}(1 - \delta/2N)$, where $\delta = 0.05$ (Bonferroni over the full pool, as in Part I). The gap is $|g_{(1)}| - |g_{(2)}|$, and $\Delta_i = |g_{(1)}| - |g_i|$.

Oracle versus measured. Steps 2–3 are oracle: exact covariances drive the designs and exact gradients fix the elimination trajectory. Their numbers are ceilings. Steps 1 and 4 sample measurement outcomes.

2 Measurement model (Step 0)

Measured groups. A context is measured by a Clifford U_α followed by readout of all n qubits. One shot therefore gives the eigenvalues of all 2^n elements of the maximal abelian group $\mathcal{G}_\alpha = \{\pm U_\alpha^\dagger Z^z U_\alpha\}$. A library Pauli P_ℓ is *measured by* α if $P_\ell \in \mathcal{G}_\alpha$, equivalently if it commutes with all n generators of $\mathcal{G}_\alpha$.

Completion. Let r_α be the rank of the parent members in $\mathbb{F}_2^{2n}$. The generators are completed to n by adding, one at a time, the first candidate that commutes with all current generators and lies outside their span. Two candidate orders are used:

- *canonical*: single-qubit Z_q , then X_q , then Y_q ;
- *mass*: library Paulis in decreasing order of $\sum_i |A_{i\ell}|$.

If a candidate list runs out, a vector of the symplectic complement is used. A seeded *random* order serves as a control. “Span” (no completion) is a diagnostic lower bound on overlap, not a circuit.

Circuit synthesis. The circuit is built from the binary tableau $[X|Z]$ of the n generators in four steps:

1. Row-reduce X , and apply H on the non-pivot qubits, which makes X invertible.
2. Row-reduce to $X = I$. This forces Z to be symmetric.
3. Apply S on every qubit with $Z_{qq} = 1$, and CZ on every pair with $Z_{ab} = 1$.
4. Apply H on all qubits.

Signs are not tracked during synthesis. Instead, every Pauli of interest is propagated through the gate list with exact phases ($P = i^k X^x Z^z$), which gives $U_\alpha P_\ell U_\alpha^\dagger = s_\ell Z^z$ directly.

Moments and sampling. With outcome distribution $p_\alpha(b) = |\langle b|U_\alpha|\psi\rangle|^2$, all group means come from one Walsh–Hadamard transform:

$$E_\alpha[\mathbf{z}] = \sum_b p_\alpha(b)(-1)^{\mathbf{z} \cdot \mathbf{b}}, \quad \mu_\ell = s_\ell E_\alpha[\mathbf{z}_\ell], \quad \text{Cov}(P_l, P_m) = s_l s_m (E_\alpha[\mathbf{z}_l \oplus \mathbf{z}_m] - E_\alpha[\mathbf{z}_l]E_\alpha[\mathbf{z}_m]). \quad (1)$$

The exact p_α gives the oracle moments. A histogram of m sampled outcomes gives the empirical ones, with the factor $m/(m-1)$ on the covariance. Sampling m shots is a single multinomial draw over 2^n outcomes, whose cost does not depend on m .

3 Estimators and levels

For a fixed context library, every linear unbiased estimator has the form

$$\hat{g}_i = \sum_\alpha \mathbf{x}_{i\alpha}^\top \hat{\mu}_\alpha, \quad \sum_{\alpha: P_\ell \in \mathcal{G}_\alpha} x_{i\alpha\ell} = A_{i\ell} \quad \forall \ell. \quad (2)$$

Let $M_q = F_{i\alpha} = \sum_\ell x_{i\alpha\ell} P_\ell$ be the fragments, let C hold their coefficients, and let B select each generator’s fragments. Then Eq. (2) is exactly $A = BC$. Every design is checked for $\max |A - BC| \leq 10^{-9} \max |A|$ and saved as the pair (B, C) .

The levels differ only in which coordinates (α, ℓ) may be non-zero:

- II-0** $\ell \in \text{supp}(C_i)$ in its Part I parent context only. This is Part I’s estimator; nothing is free.
- II-A** $\ell \in \text{supp}(C_i)$ in every context that measures it (coefficient splitting).
- II-B** II-A, plus Paulis with $A_{i\ell} = 0$ that are measured by at least two contexts already carrying C_i (zero-sum ghosts). Candidates come from the rest of the library, i.e. other generators’ supports. They are capped at 200 per generator, ranked by the single-context control-variate score $\sum_\alpha \text{Cov}(F_{i\alpha}^{(0)}, P_\ell)^2 / \text{Var } P_\ell$, where $F^{(0)}$ is the II-0 fragment.

Coefficients are post-processing. Any design satisfying Eq. (2) can be applied to every shot already taken.

4 Allocation, trajectories and cost metrics

Allocation. With n_α shots per context, $\text{Var } \hat{g}_i = \sum_\alpha q_{i\alpha}/n_\alpha$, where $q_{i\alpha} = \mathbf{x}_{i\alpha}^\top \Sigma_\alpha \mathbf{x}_{i\alpha}$ is the one-shot fragment variance. The allocation solves

$$\min_{\mathbf{n}} \sum_\alpha n_\alpha \quad \text{s.t.} \quad \text{Var } \hat{g}_i \leq \epsilon^2 \quad \forall i \in \mathcal{A}, \quad n_\alpha \geq c_\alpha. \quad (3)$$

- With $c = 0$ this is Part I’s solver, reused unchanged.
- With committed shots $c \neq 0$ (“top-up”) it is solved by multiplicative dual ascent. The minimiser of the Lagrangian is $n_\alpha = \max(c_\alpha, \sqrt{\sum_i \lambda_i q_{i\alpha}})$, updated by $\lambda_i \leftarrow \lambda_i (\text{Var } \hat{g}_i / \epsilon^2)^{1/2}$. A final one-dimensional search scales the result to the smallest feasible allocation.

Cost metrics. All three reuse Part I’s trajectories.

M1 One allocation over all N arms at $r = \text{gap}/2$.

M3 bound Part I’s exact-limit M3. The breakpoints are the distinct Δ_i ; at each, arms with $\Delta_i \leq 2r$ are active and resolved to the common radius $r = \Delta/2$.

M3 actual Part I’s event-driven M3. An arm is eliminated when $|g_j| - z\sqrt{V_j} > |g_i| + z\sqrt{V_i}$ on its *actual* variance, with each event found by bisection. This is the primary metric.

Accounting over breakpoints.

- *Maxima* (Part I): each breakpoint is solved from scratch, and each context is charged its largest request.
- *Top-up*: each breakpoint solves Eq. (3) with c equal to the shots already committed.

Designs along a trajectory.

- *Static*: one design, optimised for the full pool, is kept throughout.
- *Redesign* (oracle II-D): the design is re-optimised for the active set at every breakpoint.

5 Joint design solver (Step 3)

Problem. For an active set $\mathcal{A}$ the design problem is

$$\min_{\mathbf{x}, \mathbf{n}} \sum_{\alpha} n_{\alpha} \quad \text{s.t.} \quad \sum_{\alpha} q_{i\alpha}(\mathbf{x})/n_{\alpha} \leq \epsilon^2 \quad (i \in \mathcal{A}), \quad \text{Eq. (2)}.$$

It is jointly convex, since each term is quadratic-over-linear. Because it is scale-free, the optimal $\mathbf{x}$ does not depend on ϵ .

Why not alternate. One could alternate exact per-generator designs at fixed $\mathbf{n}$ (a linear solve over the free directions) with Part I's allocation. This is iterative coefficient splitting (ICS), and it is monotone but stalls. It is iteratively reweighted least squares on a sum of norms: once a context reaches $n_{\alpha} = 0$, no coefficient can return to it. On the winner of H₄ side 1.0 HF, ICS ends at 2.590 after 2,000 iterations, against an optimum of 2.3766 (in units of ϵ^{-2}).

Saddle form. Eliminating $\mathbf{n}$ gives

$$\min_{\mathbf{x}} \max_{\lambda \geq 0} \sum_{\alpha} h(Q_{\alpha}) - \epsilon^2 \sum_i \lambda_i, \quad Q_{\alpha} = \sum_i \lambda_i q_{i\alpha}, \quad h(Q) = n + Q/n, \quad n = \max(c_{\alpha}, \sqrt{Q}).$$

It is solved with two nested loops:

- *Inner:* L-BFGS (memory 10, at most 300 iterations) over the free coefficients of all active arms at once. The objective is smoothed by $Q \rightarrow Q + \delta^2$, with δ reduced by a factor 0.3 per outer iteration, from 0.3 to 10^{-7} of the typical context norm.
- *Outer:* 20 multiplicative dual steps per iteration. The loop stops when the exact cost has not improved by 10^{-7} (relative) for 8 iterations.

The reported cost is always the exact allocation of the returned design, so it is feasible whether or not the solver has converged.

Validation.

- The single-generator optimum matches an independent smoothed L-BFGS minimisation to 10^{-4} .
- Small joint instances match SLSQP to 2×10^{-4} .
- The analytic two-context ghost example is reproduced to 10^{-9} .
- Optimised fragment variances match a state-vector computation to 10^{-10} .

On the large problems the solver's optimality is not certified.

6 Online experiment (Steps 1 and 4)

Loop. This is Part I's successive elimination, starting at $r = \max_i |g_i|$ with $r \leftarrow \gamma r$ each round and stopping when one arm remains or $r < 10^{-3} \max_i |g_i|$. Each round does the following:

1. Allocate shots for the active arms at $\epsilon = r/z_{\delta}$ (maxima or top-up), rounded up to integers.
2. Draw one multinomial histogram per context for the new shots.
3. Estimate $\hat{g}$ from the sample means.
4. Eliminate arms, by one of two rules:
 - *marginal:* Part I's Eq. (4);
 - *pairwise:* i is dropped if some j has $|\hat{g}_j| - |\hat{g}_i| > z_{\delta} \sqrt{V_{ij}}$, where $V_{ij} = \text{Var}(\hat{g}_i - s_{ij} \hat{g}_j)$ and $s_{ij} = \text{sign}(\hat{g}_i \hat{g}_j)$.

Every context that carries a fragment receives at least one shot, because deterministic outcomes still have to be observed once. In Step 1 the radii use the oracle fragment (co)variances, as Part I does.

Learned designs (Step 4). The loop above is run with II-A coefficients learned from the data, as follows.

- *Covariance model.* Per context and fold, $\tilde{\Sigma} = \lambda \Sigma^{\text{prior}} + (1 - \lambda) \hat{\Sigma}^{\text{emp}}$ with $\lambda = \nu/(\nu + m)$, where m is the number of shots. $\hat{\Sigma}^{\text{emp}}$ is the unbiased sample covariance of Eq. (1). It is positive semidefinite, and so is every shrinkage of it, so no projection is needed. The priors are the HF determinant (in the CISD problem's own orbital basis), none ($\nu = 0$), or the oracle (a diagnostic).

- *Cross-fitting.* Every context’s shots are split into two folds. The design fitted on fold A (an exact per-generator solve at fold A’s shots) is applied to fold B’s means, and vice versa. The two estimates are averaged. Each half-estimate is unbiased whatever its design.
- *Minimum data.* A context may carry learned coefficients only when both folds hold at least 50 shots in it. Elsewhere a generator keeps its II-0 coefficients.
- *Guard.* A fitted design is kept for generator i only if the other fold’s covariance predicts a lower variance than the II-0 coefficients do. Otherwise i falls back to II-0.
- *Schedule.* The first round is an II-0 pilot. Designs are refitted whenever the shots held have grown by a factor of 1.5.
- *Radii.* Either estimated or oracle (the latter to isolate the design effect). Estimated radii use only the *held-out* fold’s covariance: design $\mathbf{x}^f$ is evaluated with $\tilde{\Sigma}^{\bar{f}}$, which is independent of it. For the cross-fitted estimate,

$$\text{Cov}(\hat{g}_i, \hat{g}_j) = \frac{1}{4} \sum_f \sum_{\alpha} \mathbf{x}_{i\alpha}^{f\top} \tilde{\Sigma}_{\alpha}^{\bar{f}} \mathbf{x}_{j\alpha}^f / n_{\alpha}^{\bar{f}},$$

where $\bar{f}$ is the fold not used to fit $\mathbf{x}^f$.

- *Allocation.* Shots are planned with exactly the per-shot variance the radii will use, $\frac{1}{2} \sum_f \mathbf{x}_{i\alpha}^{f\top} \tilde{\Sigma}_{\alpha}^{\bar{f}} \mathbf{x}_{i\alpha}^f$ per context, and topped up from the shots already held.

Three failures of the first Step 4 implementation. Each was found on LiH or H₄ side 2.0 and is now covered by a test or a diagnostic.

1. *Learning from too little data.* Fitting straight after the pilot round used covariances from one or two samples per fold. Their spurious zero-variance directions were exploited by the design and the radii alike. On LiH, 46 of 100 trials eliminated almost every arm in round 2, and only 54% selected the winner. Fix: the minimum-data rule.
2. *Radii from a covariance that includes the fitting fold.* A design’s variance was estimated from both folds, one of which it was fitted to. Fix: held-out radii. A test checks that the estimated variance matches the actual spread of the estimate across replicates (0.8–1.35).
3. *Planning and radii that disagreed.* Planning used one fold’s design and covariance, while the radii used both. On H₄ side 2.0, a context whose 5-shot fold happened to show zero variance was never topped up, while the other fold kept a variance there that the radii could not shrink. Two trials ran to the radius floor, at 7.8×10^8 shots each, against a median of 1.3×10^4 . Fix: plan with the radii’s own variance.

7 Results

7.1 Step 0: validation

- Circuits map all 2^n elements of random maximal groups ($n \leq 5$) to $\pm Z^z$, and conjugation phases agree with dense matrices.
- Group means and covariances match Part I’s state-vector evaluator, and II-0 fragment covariances match Part I’s context covariances, to 10^{-12} .

Table 1 checks the sampler against the analytic covariance.

7.2 Step 1: II-0 as a shot-level experiment

Gates.

- *Noiseless.* With every estimate set to the exact gradient and real-valued shot counts, the online loop reproduces Part I’s zero-noise trial exactly: ratio 1.0000000 on all nine case/rule pairs. With integer shots the difference is at most 0.08% on H₄ and 1.8% on LiH, where rounding up drops one arm a round early.
- *Sampled.* At $\gamma = 0.5$ with maxima accounting (250 trials; 100 on LiH), the shot-level runs agree with Part I’s Gaussian surrogate within 1.5 standard errors on every row (Table 2).

All Step 1 and Step 4 runs use per-trial random streams: trial i uses `SeedSequence(seed).spawn(n)[i]`. Results therefore do not depend on how the trials are split across processes or machines.

Table 1: Shot-level sampler at scale $\times$ Part I’s M1 allocation; 2,000 replicates on H₄ and 400 on LiH. Columns: largest bias of any $\hat{g}_i$ in standard errors; largest $|\widehat{\text{Var}}/\text{Var} - 1|$ (tolerance $5\sqrt{2/R} = 0.158$ and 0.354); largest correlation error (tolerance $5/\sqrt{R} = 0.112$ and 0.250); two-sample KS p against Part I’s Gaussian surrogate for the winner and for the binding generator. All rows pass. The small p for the H₄ 2.0 HF winner (0.003) reflects the lattice of outcomes on a determinant, not a bias.

Case	scale	shots	bias (σ)	var. err.	corr. err.	KS winner	KS binding
H ₄ 1.0 HF	1	2,244,893	2.56	0.066	0.060	0.79	0.21
H ₄ 1.0 HF	0.001	2,273	1.81	0.051	0.070	0.35	0.64
H ₄ 1.0 CISD	1	59,671	1.87	0.078	0.075	0.86	0.86
H ₄ 1.0 CISD	0.001	88	2.26	0.080	0.064	0.86	0.86
H ₄ 2.0 HF	1	598,129	1.97	0.049	0.074	0.003	0.37
H ₄ 2.0 HF	0.001	628	1.91	0.080	0.069	0.03	0.41
H ₄ 2.0 CISD	1	58,094	1.82	0.109	0.065	0.20	0.35
H ₄ 2.0 CISD	0.001	86	2.79	0.059	0.074	0.53	0.33
LiH 3.0 HF	1	3,564,398	3.53	0.140	0.190	0.32	0.52

Table 2: Mean cost of Part I’s surrogate against Part II’s shot-level loop, $\gamma = 0.5$, maxima accounting, with the difference in combined standard errors. Part I’s correlated model is unavailable for LiH (too many generator pairs); its independent-noise marginal run is compared.

Case	marginal rule: Part I / Part II	pairwise rule: Part I / Part II
H ₄ 1.0 HF	1,282,876 / 1,315,987 (+0.5)	697,109 / 874,589 (+1.1)
H ₄ 1.0 CISD	151,337 / 150,437 (−0.3)	89,807 / 95,256 (+1.5)
H ₄ 2.0 HF	534,126 / 543,563 (+0.5)	507,751 / 522,126 (+0.1)
H ₄ 2.0 CISD	102,018 / 102,348 (+0.1)	51,726 / 51,059 (−0.3)
LiH 3.0 HF	880,804 / 856,976 (−0.7)	—

Table 3: II-0 with the pairwise rule, shot-level sampling and oracle radii; 250 trials (LiH 100). “Part I M3 as run” is the marginal rule, $\gamma = 0.5$, maxima accounting (mean / median). The **bold** column is the Part II baseline used from here on. Entries are mean $\pm$ s.e.m., with median, 90th percentile and number of wrong selections for the baseline.

Part I M3 as run		$\gamma = 0.5$		$\gamma = 0.9$			
Case	mean / median	maxima	top-up	maxima	top-up	median / P90	fail
H ₄ 1.0 HF	1,315,987 / 921,024	874,589	782,213	406,172	385,766 ± 53,637	285,250 / 568,386	2
H ₄ 1.0 CISD	150,437 / 163,910	95,256	96,150	43,218	41,515 ± 921	39,754 / 61,289	0
H ₄ 2.0 HF [†]	543,563 / 627,922	522,126	498,755	1.60×10^6	$1.62 \times 10^6 \pm 0.83 \times 10^6$	160,459 / 295,264	3
H ₄ 2.0 CISD	102,348 / 128,284	51,059	50,145	23,748	23,827 ± 613	22,136 / 37,671	0
LiH 3.0 HF	856,976 / 784,940	582,681	580,624	317,785	307,170 ± 6,238	299,479 / 395,948	0

[†]H₄ 2.0 HF has an exactly degenerate runner-up (Part I). A few trials never separate the tie and run to the radius floor, so the mean is dominated by them; use the median.

Baseline. Table 3 gives the II-0 baseline.

Reading the table.

- On medians, the baseline is 62–83% cheaper than Part I’s M3 as run. Most of that comes from Part I’s own two findings: the pairwise rule and a finer radius schedule.
- The price is a small error rate on the HF rows: 5 wrong selections in 1,100 trials, all on H₄ HF, where the leading generators are exactly or nearly tied. The CISD rows and LiH had none, and Part I’s marginal rule had none. This is still far below the nominal $\delta = 0.05$.
- Top-up and maxima accounting cost about the same here. An earlier run, with a different random stream, suggested that top-up removes the H₄ 2.0 HF tail; the per-trial reruns do not confirm it.

7.3 Step 2: overlap census

Overlap is abundant, and most of it is present before any completion. The binding generator is 89–100% splittable on LiH and H₂O. Coefficient splitting (II-A) passes the go/no-go test.

Table 4: Multiplicity (*mult.*): mean number of contexts whose measured group contains a Pauli of $\mathcal{B}_0$. *Split.*: share of $\mathcal{B}_0$ with multiplicity ≥ 2 . *Mass*: share of $\sum |A|$ on those Paulis for the pool, the winner and the binding generator (the one that sets M1’s cost). CZ: two-qubit gates per measurement circuit. Aux: mean number of II-B candidates per generator before the cap. Supports do not depend on the state, so HF and CISD differ only through the winner and binding generator.

Case	span		canonical		mass completion							aux
	mult.	split.	mult.	split.	mult.	split.	pool	winner	binding	CZ		
H ₄ 1.0 HF	1.71	52%	1.88	58%	1.95	60%	62%	61%	66%	8.4	386	
H ₄ 1.0 CISD	1.71	52%	1.88	58%	1.95	60%	62%	67%	67%	8.4	386	
H ₄ 2.0 HF	1.71	52%	1.88	58%	1.95	62%	65%	71%	61%	8.2	379	
H ₄ 2.0 CISD	1.71	52%	1.88	58%	1.96	62%	65%	68%	87%	8.2	380	
LiH 3.0 HF	2.57	67%	2.73	71%	2.95	76%	84%	87%	89%	23.3	19,348	
H ₂ O eq HF/CISD	2.72	67%	2.89	70%	3.10	74%	89%	88%	99%	32.2	50,703	
H ₂ O str HF/CISD	2.72	67%	2.88	70%	3.11	75%	88%	87%	100%	32.3	50,527	

7.4 Step 3: oracle ceiling

Table 5 gives every metric in context-shots. Percentages are relative to II-0 on the same metric; redesign is compared with II-0 under top-up accounting.

Table 5: Oracle ceilings. *Static* designs are optimised for the full pool; *redesign* re-optimises at every breakpoint (top-up accounting). Redesign was run on H₄ only; for H₂O only II-A with the mass completion was run.

Case	compl.	level	M1	M3 bound (max.)	M3 bound (top-up)	M3 actual	redesign (top-up)
H ₄ 1.0 HF	–	II-0	2,244,862	566,950	566,534	569,020	–
	mass	II-A	719,905	358,865	351,942	352,830 (–38%)	241,899 (–57%)
	mass	II-B	671,127	343,891	338,007	339,202 (–40%)	239,617 (–58%)
	canonical	II-A	780,960	382,898	369,746	375,424 (–34%)	293,701 (–48%)
	canonical	II-B	690,112	369,190	356,615	358,077 (–37%)	290,326 (–49%)
H ₄ 1.0 CISD	–	II-0	59,641	53,752	49,327	49,729	–
	mass	II-A	25,497	24,652	23,589	23,102 (–54%)	23,093 (–53%)
	mass	II-B	23,937	23,090	22,274	22,037 (–56%)	21,828 (–56%)
	canonical	II-A	30,796	29,926	27,130	27,745 (–44%)	26,777 (–46%)
	canonical	II-B	28,427	27,599	25,237	25,555 (–49%)	24,981 (–49%)
H ₄ 2.0 HF	–	II-0	598,102	220,388	220,311	228,733	–
	mass	II-A	240,381	127,470	127,432	136,685 (–40%)	89,779 (–59%)
	mass	II-B	162,830	111,606	111,581	113,507 (–50%)	88,597 (–60%)
	canonical	II-A	263,473	150,010	149,048	157,123 (–31%)	92,049 (–58%)
	canonical	II-B	167,208	119,818	119,131	123,328 (–46%)	90,942 (–59%)
H ₄ 2.0 CISD	–	II-0	58,067	33,385	31,528	31,553	–
	mass	II-A	28,023	18,594	18,145	17,978 (–43%)	15,754 (–50%)
	mass	II-B	22,966	16,276	16,008	15,869 (–50%)	13,579 (–57%)
	canonical	II-A	29,169	18,493	18,034	18,041 (–43%)	15,605 (–51%)
	canonical	II-B	24,778	16,445	16,189	16,191 (–49%)	13,415 (–57%)
LiH 3.0 HF	–	II-0	3,563,904	637,633	586,151	333,272	–
	mass	II-A	581,663	118,386	107,085	98,644 (–70%)	–
	mass	II-B	580,654	117,027	105,489	98,998 (–70%)	–
	canonical	II-A	723,831	146,184	132,268	81,027 (–76%)	–
	canonical	II-B	714,750	141,616	127,460	121,174 (–64%)	–
H ₂ O eq HF	–	II-0	1.72×10^8	2.21×10^7	2.16×10^7	1.32×10^7	–
	mass	II-A	2.71×10^7	4.19×10^6	3.85×10^6	2.69×10^6 (–80%)	–
H ₂ O eq CISD	–	II-0	4.12×10^{11}	4.58×10^{10}	4.10×10^{10}	2.44×10^{10}	–
	mass	II-A	8.50×10^{10}	9.89×10^9	8.51×10^9	6.05×10^9 (–75%)	–
H ₂ O str HF	–	II-0	3.86×10^9	2.02×10^7	1.89×10^7	1.46×10^7	–
	mass	II-A	6.05×10^8	5.85×10^6	5.52×10^6	5.25×10^6 (–64%)	–
H ₂ O str CISD	–	II-0	5.72×10^9	1.71×10^8	1.56×10^8	1.22×10^8	–
	mass	II-A	1.54×10^9	5.35×10^7	4.59×10^7	3.89×10^7 (–68%)	–

Observations.

- *Every level passes the 10% gate on every case.* On M3 actual, II-A reduces the cost by 38–54% on H₄, 70–76% on LiH and 64–80% on H₂O.
- *Ghosts add little.* II-B adds 2–10 points over II-A on H₄, and none on LiH with the mass completion. With the canonical completion on LiH it is 12 points worse on M3 actual while better on M1. The static design optimises the full-pool (M1-shape) objective, so downstream M3 costs need not be monotone in the level.
- *Completion matters modestly.* On H₄, mass is 0–9 points better than canonical. On LiH, canonical is 5 points better for II-A.
- *Redesign needs top-up accounting.* On the M3 top-up bound, redesign improves the depth-dominated H₄ HF rows from −38%/−42% (static) to −57%/−59%, and the CISD rows by 1–7 points. Under maxima accounting it can be worse than static: on H₄ 1.0 CISD, 31,447 against 24,652.
- *Top-up alone helps.* It lowers the II-0 bound by 0–11%, which closes Part I’s open item “allocate incrementally”.

Prior transfer. At fixed geometry, HF and CISD share one context library. A design fitted on the HF-determinant covariances can therefore be evaluated on the CISD state (Table 6).

Table 6: II-A (mass) designs evaluated on the CISD state, relative to II-0.

Geometry	design fitted on	M1	M3 bound	M3 actual
H ₄ side 1.0	CISD (oracle)	25,497 (−57%)	24,652 (−54%)	23,102 (−54%)
	HF, applied to CISD	93,004 (+56%)	89,355 (+66%)	83,940 (+69%)
H ₄ side 2.0	CISD (oracle)	28,023 (−52%)	18,594 (−44%)	17,978 (−43%)
	HF, applied to CISD	83,159 (+43%)	55,577 (+66%)	52,383 (+66%)

On a determinant, outcomes are either deterministic or perfectly (anti)correlated. A design fitted to that structure relies on cancellations the correlated state does not have.

7.5 Step 4: designs learned from the shots

Setup.

- *Cases.* The two H₄ CISD rows, with the HF determinant as the classical prior (200 trials each), and LiH (100 trials).
- *LiH prior.* LiH’s truth *is* its HF determinant, so an HF prior would be the oracle. A flat prior (unit variances, no correlation) replaces it there.
- *Common settings.* II-A with the mass completion, the pairwise rule, $\gamma = 0.9$, top-up allocation and shot-level sampling, all in the loop of Sec. 6. Seed 2.
- *Comparisons.* Changes are relative to II-0 with the same kind of radii.

Table 7: Learned designs. Mean (change) and correct selections, out of 200 on H₄ and 100 on LiH. Estimated radii make a run fully non-oracle; “oracle” rows use exact covariances for the radii, and “II-A oracle” also for the design. Medians are in the text where the mean is tail-dominated.

Design	radii	H ₄ 1.0 CISD		H ₄ 2.0 CISD		LiH 3.0 HF	
		mean (change)	ok	mean (change)	ok	mean (change)	ok
II-0	oracle	42,313	200	25,563	200	305,794	100
II-A, oracle covariance	oracle	24,220 (−43%)	200	11,516 (−55%)	200	146,188 (−52%)	100
II-A, HF prior only	oracle	1.19×10^9	200	7.39×10^8	194	—	—
II-0	estimated	42,314	200	24,123	200	303,118	100
II-A, HF prior only ($\nu = \infty$)	estimated	81,047 (+92%)	197	143,851 (+496%)	171	—	—
II-A, HF shrunk, $\nu = 1000$	estimated	32,099 (−24%)	200	31,696 (+31%)	195	—	—
II-A, HF shrunk, $\nu = 100$	estimated	28,307 (−33%)	200	15,293 (−37%)	200	—	—
II-A, flat prior, $\nu = 100$	estimated	—	—	—	—	193,042 (−36%)	100
II-A, data only ($\nu = 0$)	estimated	27,129 (−36%)	200	13,727 (−43%)	200	184,511 (−39%)	100
II-A, data only, no guard	estimated	26,749 (−37%)	200	13,764 (−43%)	200	184,511 (−39%)	100

Findings.

- *Learning from the data recovers 75–84% of the ceiling, fully non-oracle, with no wrong selections.* II-A with covariances estimated from the shots costs 36%, 43% and 39% less than II-0 run the same way. The online oracle ceiling is 43%, 55% and 52%. All 500 trials selected the winner. The data-only rows have no tail: the 90th percentiles are 36,052, 20,382 and 228,603.
- *Estimating variances is nearly free at II-0.* Estimated and oracle radii differ by at most 6% (within noise), with no wrong selections.
- *The HF prior alone fails twice over.*
 - Its design is wrong for the correlated state.
 - Its near-zero variances corrupt the radii (85–99% correct, heavy tails; medians 32,540 and 25,448).
 - They also corrupt the allocation: with oracle radii, runs stall at the radius floor (10^8 – 10^9 shots).
- *A prior helps only with little weight, and never beat the data alone here.* With the HF prior, $\nu = 100$ is 3–6 points worse than data alone, and $\nu = 1000$ is worse still, with 5 wrong selections on side 2.0. On LiH the flat prior at $\nu = 100$ is 3 points worse.
- *The guard rarely acts once the minimum-data rule is in place.* With and without it the results agree within noise. It is kept as a safety net. Its apparent importance in an earlier run came from the planning mismatch (failure 3 in Sec. 6).

Caveats.

- The guard chooses between two unbiased estimators using the evaluation fold’s second moments, which could bias the estimate at second order. A fixed-shot test finds no bias above 4σ over 150 replicates, and estimated variances within 0.8–1.35 of the actual spread.
- Designs are refitted per generator at fixed shots, not with the joint solver, so the learned costs are upper bounds on what a jointly learned design would achieve.
- The minimum-data threshold (50 shots per fold) is a single, untuned choice.
- H₂O has not been run in Step 4 yet; see Sec. 9 for sizing.

8 Conclusions and next steps

1. The shot-level II-0 baseline (pairwise rule, $\gamma = 0.9$, top-up) is 62–83% cheaper than Part I’s M3 as run (medians), and it is the reference for every Part II level. It made 5 wrong selections in 1,100 trials, all on the tied H₄ HF rows.
2. Overlap is abundant, and the oracle ceiling of coefficient splitting is large: 38–80% on M3 actual across the cases computed.
3. Ghost Pauli products add little. Per-breakpoint redesign pays on depth-dominated rows, but only with top-up accounting.
4. The Hartree–Fock covariance is harmful as a design prior.
5. Learned from the shots, II-A keeps 75–84% of its ceiling without any oracle input: –36%, –43% and –39% against II-0 on the two H₄ CISD rows and LiH, with no wrong selections and no cost tail. It needs three safeguards:
 - cross-fitting;
 - radii and allocation computed on the held-out fold;
 - a minimum number of shots per fold before a context is learned.

The default should be data-driven covariances. A prior gave no benefit here and should never be used alone.

Next.

1. *Broaden Step 4.* Run it on the H₂O CISD rows (cluster; Sec. 9). Swap the per-generator refit for the joint solver, and tune the minimum-data threshold.
2. *Learned redesign (II-D).* Refit for the active set at each elimination, with top-up accounting, on the depth-dominated rows, where the oracle adds up to 19 points.
3. *Correlated trajectory states.* Replace the CISD proxy with saved ADAPT trajectory states (Part I’s open item), and study the prior weight ν there.

4. *Price the circuits.* The mass completion uses 8–32 CZ gates per context. Compare against a completion constrained by circuit depth.
5. *Leave II-B, II-C and II-E for later.* Ghost Pauli products added little at the oracle level.

9 Reproduction

Code. `part2_estimator_design/` on branch `part2`. It imports Part I’s `src/` directly and builds problems into its own `.cache/`.

Environment. WSL Ubuntu, Python 3.12.3, with the versions pinned in `requirements.txt`: NumPy 2.5.2, SciPy 1.18.1, PySCF 2.14.0, OpenFermion 1.8.1. NumPy ≥ 2 is required. `part2_estimator_design/Dockerfile` builds the same environment from the repository root. It is not yet tested, since no Docker was available here. On a cluster, either convert it with Apptainer or create a venv from `requirements.txt`.

Modules (`src/`):

- `symplectic, clifford`: Pauli algebra, completion and circuits;
- `contexts`: context library and census;
- `sampler`: moments and sampling;
- `design`: levels and the joint solver;
- `allocation`: top-up allocation;
- `planning`: M1 and M3 trajectories;
- `online`: Step 1 loop;
- `learning`: Step 4;
- `parallel`: trial parallelism and sharding.

Commands (from `part2_estimator_design/`; set `OMP_NUM_THREADS=1` when using `--workers`):

```
python scripts/build_problems.py                # 9 cases, checked against Part I
python -m pytest tests -q                        # 53 tests
python scripts/step0_validate.py --low-shot 0.001 # H4 sampler gate
python scripts/step0_validate.py --cases LiH_R3p0_HF --replicates 400 \
--out runs/step0_lih
python scripts/step1_online_baseline.py --trials 250 --workers 4 --cases <H4 cases>
python scripts/step1_online_baseline.py --trials 100 --workers 8 --cases LiH_R3p0_HF
python scripts/step1_online_baseline.py --gates-only # exact noiseless gate
python scripts/step2_overlap_census.py           # all nine cases
python scripts/step3_oracle_ceiling.py --cases <H4 cases> --redesign
python scripts/step3_oracle_ceiling.py --cases LiH_R3p0_HF \
--strategies mass canonical
python scripts/step3_oracle_ceiling.py --cases <H20 cases> --strategies mass \
--levels II-0 II-A
python scripts/step3_prior_transfer.py
python scripts/step4_learned_designs.py --trials 200 --workers 4 # H4 CISD rows
python scripts/step4_learned_designs.py --cases LiH_R3p0_HF --trials 100 --workers 10
python scripts/aggregate_results.py
```

For a cluster, split any Step 1 or Step 4 run with `--shard K/N`, one job per `K`. Then run `scripts/merge_trials.py --step step4 --cases <case>`. It checks that every trial is present exactly once and writes the same summary file as an unsharded run.

Outputs. Each script writes `runs/<case>/<case>_<step>.csv`, with a `*_meta.json` recording package versions and the git commit. Steps 1 and 4 also keep every trial in `runs/<case>/trials/`. Step 3 designs are saved as `*_B.npz` and `*_C.npz`; these are not tracked and are regenerated by the script.

Seeds.

- Step 0: seed 0.
- Step 1: seed 1 for Part II, and seed 0 for Part I's surrogate.
- Step 4: seed 2.

In Steps 1 and 4, trial i uses `SeedSequence(seed).spawn(n)[i]`, so results do not depend on `--workers` or `--shard` (tested).

Runtimes and memory (one desktop, 8 cores/16 threads, 15 GB):

- Tests: about 3 min.
- Step 3, H_4 with redesign: 15–20 min per case. LiH: 5–15 min per design. H_2O : 12–47 min per design.
- Step 4, H_4 : 2–5 min per configuration (200 trials, 4 workers). The prior-only configuration with oracle radii takes 8–25 min, because its runs stall until the radius floor.
- Step 4, LiH: 13–24 min per configuration (100 trials, 10 workers, about 1.5 GB per worker). A sparse direct solve in the design refit made LiH $8\times$ faster than the original conjugate-gradient solve.
- Step 4, H_2O eq. CISD, one trial: II-0 takes 329 s at a 3.6 GB peak, and II-A (data only) 685 s at a 7.1 GB peak. Five configurations $\times$ 100 trials is therefore about 60–80 CPU-hours per H_2O case. Memory is the constraint, at about 7 GB per process. For example, a 64-core, 256 GB node runs about 32 processes and finishes one case in roughly 5 hours. Plan the cluster jobs as `--shard K/N` with `--workers` set by memory.